Risk List, Feasibility Analysis & Mitigation Plan

Originally an Inception deliverable; updated during Elaboration Iteration 1 (mitigation strategies added) and Iteration 2 (progress status added)

Risk Register

ID	Category	Risk	Likelihood	Impact	Mitigation Strategy	Status (Iteration 2)
R1	Technical	Lack of experience with POS system architecture.	Medium	High	Base the design on an established reference architecture (Larman's POS case study, layered design) rather than designing from scratch; allocate research time in early Elaboration.	**Mitigated** — layered architecture (UI / Domain / Business Infrastructure / Technical Services) modeled on the reference case study, completed in Iteration 1.
R2	Technical	Integration challenges between modules (sales, inventory, reporting).	Medium	Medium	Define clear layer boundaries and a single interface contract (REST API) between the UI and backend before any coding starts.	**In Progress** — layer boundaries and interfaces defined in the architecture; end-to-end integration testing is scheduled for Construction.
R3	Technical	Data security vulnerabilities.	Medium	High	Hash/encrypt stored passwords, enforce role-based access	**Planned** — addressed in design via the Authentication/Authorization Controller; implementation happens in Construction.

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					control (NFR2), require HTTPS for all API calls, validate all input server-side.	
R4	Technical	Incorrect scanning of a product/gadget barcode.	Medium	Medium	Provide a manual product-code/name search as a fallback to barcode scanning; validate scanned codes against the product database before adding an item to a sale.	**Mitigated in design** — the Search Product and View Product Details use cases provide the manual fallback path.
R5	Business	System may not meet stakeholder expectations.	Low	High	Validate use cases and the UI prototype against the business case and marking guide at the end of every iteration.	**In Progress** — use cases and a UI prototype were produced this iteration specifically for that review.
R6	Business	Budget or infrastructure limitations.	Low	Medium	Use free/open-source tooling throughout (Python backend, React UI,	**Mitigated** — technology stack confirmed as free/open-source; no paid infrastructure required to complete the coursework.

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					free-tier hosting) instead of paid infrastructure.	

Feasibility Analysis

Technical Feasibility

The project is technically feasible using modern, freely available programming frameworks and databases (Python backend, React front end, a relational or lightweight database).

Economic Feasibility

Development costs are manageable within the academic project scope; no paid infrastructure is required to complete and demonstrate the system.

Operational Feasibility

Store staff can be trained easily due to the system's user-friendly design — reinforced by NFR3 and validated by the UI prototype built in Elaboration Iteration 2.